

Ch9 Statistical Testing: Bootstrap & Permutation

Source: reading_pack/md/A09_Bootstrap.md | Full textbook: TEXTBOOK_Stability_Identifiability.md

9 Bootstrap & Permutation

9.1 Bootstrap Stability

```
for b=1..20:  
  sample 80% male + 80% female with replacement  
   $\theta^b$  = probe( $X_b$ ,  $y_b$ )  
  Stab = mean_{i≠j} cos( $\theta^i$ ,  $\theta^j$ )
```

- 0.979 ± 0.001 male/female
- 0.979 ± 0.001 mean off-diag ± 0.001 std ± 0.001 shuffled null

9.2 Permutation PrivGap

```
null = { PrivGap_rand^1 ... PrivGap_rand^20 }  
p = (# null ≥ PrivGap_true + 1)/(20+1)
```

- $p < 0.01$
- corr(Stab, PrivGap) = 0.979 ± 0.001 Stability ± 0.001 Identifiability

** Wasserman Ch.8 Bootstrap Efron 1979

1. 2. 3. 4. 5. 6. 7. 8. 9. 10.

1. 1. 2. 3. 4. 5. 6. 7. 8. 9. 10.
 2. 1. 2. 3. 4. 5. 6. 7. 8. 9. 10.
 3. Ch6 Wang Fig.3 Kim Fig.2 Fig.2 heuristic Fig.3 optimal
 4. DEFINITIONS.md IDEA_CARD.md
- reading_pack A1_A2... PDF